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Chapter 9

Visual Analogy - a Strategy for Design Reasoning and Learning

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Abstract

This paper proposes that the use of visual analogy in problem solving is an example of similarity-based reasoning, cognitively facilitated by imagistic operations. We show that in designing, which is an example of ill-structured problem solving, this type of reasoning is most valuable. We offer evidence that novices, in particular, benefit from guidance to use analogy, which helps them to better understand abstract concepts and to fully exploit their capacity to retrieve and implement previously acquired knowledge.

1 Two Systems of Reasoning

Four decades ago the design research community in its entirety, more or less, subscribed to the notion that design practices should be made to follow exacting methods, the development of which became the single most important concern on design research agendas. Members of the so-called "design methods movement" worked hard to help develop a design science: it was seen as an important step forward in the evolution of the design disciplines and their standing among the professions. (The professions, in turn, struggled to gain an academic status that would put them on an equal footing with the sciences). Consequently, so-called 'traditional' design practices, based on 'soft' strategies including intuition and the use of informal knowledge was looked down upon. In contrast, the idea of a 'hard' analytic science of design sounded very appealing, as eloquently expressed by members of the methods movement (e.g. Gregory 66) and by Herbert Simon, in *The Sciences of the Artificial* (Simon 81).

The experience of the last few decades has taught us that we are far from attaining the goal of formulating a design science; indeed, the very goal has been submitted to serious questioning. It turns out that in most if not all design domains, rigorous design methods based on well-defined algorithms do not yield the expected improvement in design quality. In some cases, particularly in architecture and in industrial design, it is very hard to get designers in the real world to even try them out. As a result, we have been forced back to

the drafting board - and to the keyboard as well, this time - to take a hard look at how designs, and primarily design ideas, are being generated. Two interrelated questions beg to be answered: why should design methods, based on carefully articulated sets of rules, fail to deliver better designs? And why do such methods meet with considerable resistance on the part of so many design practitioners who refuse to give up their age-old 'soft' design strategies, despite their inferior status as compared to 'scientific methods'? These questions are especially important from the perspective of design education. On the one hand, we are uncomfortable with our traditional, experiential, 'trial and error' teaching traditions; on the other hand, we fall short of being able to effectively transmit a comprehensive codex of design knowledge, including procedural knowledge about how to design and how to reason about designing. It is for this reason that design reasoning has become a research topic of great importance, motivated by the hope that knowing more about natural design reasoning processes would pave the way for modes of enhancing and teaching it. Fortunately, this question falls within a more general research agenda on human cognition, in the area of reasoning and problem solving, and work in this area is useful for our quest.

For a long time psychologists took it for granted that development follows a trajectory leading from reasoning based on similarity in early childhood, to rule-based reasoning in adulthood. The latter, in this view, signifies the ability to form abstract concepts and use symbolic representation, primarily through the use of language. This belief was, and still is in some quarters, so fundamental that very few researchers have challenged it. Similarity pertains to figurative qualities, and is therefore largely visual. Fodor (75), talking about the nature of mental visual imagery, has simply stated that children think in pictures, whereas adults think in words. Katifinann (80), who thought it wrong but who found himself fighting windmills, has talked about a "sweeping linguistic imperialism" (p. 18). According to him, linguistic representation is used to reason about problems that belong to a familiar type. For novel problems he thinks imagery is enlisted, and the reason imagery is helpful in those cases is the relative freedom of imagery from rules.

In later years cognitive scientists talked about "pop-up" or bottom-up cognitive operations in reasoning, roughly parallel to similarity and example-based inductive reasoning, as more rudimentary than the more sophisticated, central executive bottom-down cognitive operations which correspond (roughly, again) to rule-based deductive reasoning.

In recent years the primacy of rule-based reasoning is no longer universally accepted as an absolute truth, and researchers are interested in the relationship between the two modes of reasoning: rule based and similarity based, in both children and adults (Sloman and Rips 98). Sloman (96) has suggested that we are endowed with two independent (but interacting) cognitive systems, each dedicated to one mode of reasoning: one associative and similarity-based, the other symbolic and rule-based. In Sloman's account, however, there still is a clear hierarchy and the associative system is seen as 'subsymbolic'. Other researchers have

advanced the view that if indeed there are two systems of reasoning, they are equally important to processes of problem solving and learning (i.e. Gentner and Medina 98). A wider discussion of this most interesting question is unfortunately beyond the scope of the present work. For our purposes here suffice it to say that we accept Gentner and Mediria's view and our arguments will proceed on the assumption that two parallel, interactive cognitive systems operate in problem solving, one underlying reasoning on the basis of rules, and the other on the basis of similarity.

To the two-fold reasoning paradigm we add an assumption of our own, which we do not attempt to prove here, as following: The strength of activation of one or the other system of reasoning in problem solving is context dependent. There are individual differences among problem-solvers (and developmental differences as well), but mostly the type of problem, and especially whether it is well-structured or ill-structured, determines which reasoning system is most instrumental for a particular instance of problem solving. Furthermore, we postulate that for ill-structured problems similarity-based reasoning is especially useful, as there are few or no rules the problem-solver can retrieve and use (one of the properties of ill-structured problems is that no means are specified for reaching a problem's end-state, or solution, from the initial state, through interim states). Since we see designing as a process of solving problems that for the most part are ill-structured, it will come as no surprise that we are particularly interested in similarity-based reasoning.

2 Visual Analogy: An Imagistic, Similarity-Based Reasoning Strategy for Ill-Structured Problem Solving

Donald Schön, who was interested in the formulation of problems as the basis for action throughout his career, pointed out decades ago (Schön 63) how powerful the transfer of concepts from one case to the next and from one field to another can be. Gentner and Medina (98) have presented rich evidence to support the claim that similarity-based reasoning is a powerful engine in the solving of many kypes of problems. Of the various kinds of similarities that may be activated in a process of mapping and transfer from one situation to another they single out analogy. Analogy, which refers to similarity between relationships, is of particular importance because in its most complete state it establishes structural commonalties, in addition to surface features that are carried over to a new situation by other types of similarity as well[1]. It is true that the carrying over of surface

[1]We would like to stress the importance of the definition of analogy as similarity between relationships, which distinguishes it from other kinds of similarity. Analogy is sometimes equated to metaphor and even confounded with it. In our view, analogy and metaphor are two very different types of similes.

features only, without a structural similarity to underpin them, may lead to a false analogy and consequently to a wrong solution to a problem. We have discussed and illustrated this problem elsewhere (Goldschmidt 95). However, we must qualify this assertion when talking about visual analogy, where transfer of features that may be superficial otherwise, like form and shape, are actually the essential core of a situation. It is therefore important to exercise caution when determining which properties of a problem are structural and which are superficial, as their definition may vary according to the type of problem. Design problems are often concerned with form giving and therefore this question is of particular relevance to us. Gentner and Medina (98) have also shown that the use of analogy, especially by novices, contributes to the learning of new abstract concepts and to the extension and application of previously acquired knowledge. We will return to the question of learning later.

If the use of analogy does indeed facilitate problem solving of a certain type, and we have assumed that these problems are primarily ill-structured, then we must address the hitherto unanswered question: what are the underlying cognitive mechanisms that are involved in the generation of analogy, and visual analogy in particular? An answer to this question may help explain why analogy is rarely used spontaneously, as suggested by the results of several studies (e.g. Beveridge and Parkins 87; Gick and Holyoak 80; 83. Most studies styled their experiments after Duricker's famous radiation problem (Duncker 45)).[2] It may also help decide if and when it makes sense to induce the use of analogy as a problem solving strategy, especially in educational settings. We can treat these questions only in a preliminary and partial manner in this paper, and we hope that further research will make it possible to elaborate on these issues. The cognitive mechanism that we hold 'responsible' for visual analogy, which is most instrumental in design problem solving, is mental visual imagery. In the following sections we discuss the relationship between visual analogy and imagery in creative problem solving as we understand it, to be followed, in section 4, by commentary on spontaneity or lack thereof, and on educational benefits ensued from the use of guided analogy as a design strategy.

2.1 Kekulé's Insight

"...Again the atoms were gamboling before my eyes. This time the smaller groups kept modestly to the background. My mental eye, rendered more acute by repeated

[2] In this problem subjects were asked to suggest an appropriate medical procedure for a patient who suffers from a tumer in his stomach. The tumer is not operable, but a certain kind of radiation can destroy it; the problem is that at the high intensity that is required, helthy tissue around the tumer would also be destroyed. Before they were given this problem, subjects were told a story about a general who wanted to capture a fortress; this was not possible in a frontal attack because of mined roads; the solution was to divide the force and have the soldiers attack in small groups from several directions at once.

> vision of this kind, could now distinguish larger structures, of manifold conformation; long rows, sometimes more closely fitted together; all twining and twisting in snakelike motion. But look! What was that? One of the snakes had seized hold of its own tail, and the form whirled mockingly before my eyes. As if by a flash of lightning I awoke..."

This well-known citation, attributed to the 19th century chemist Friedrich Kekulé (e.g. Koestler 64, 118; Rothenberg 79, 396), is an introspective account of what went on in his mind when he had an insight that eventually led to the discovery of the ring structure of benzene (see Figure 1). The authenticity of the actual event is questionable: Kekulé made the famous comments about his insight decades after the event, in circumstances that suggest that he may have wished to be entertaining rather than factually accurate (Gruber 95). The fact that the insight allegedly occurred while Kekulé was half-asleep or at least engaged in day dreaming also adds to doubts about the scientist's ability to give a credible account of his thinking. Some critics go as far as to argue that the entire story amounts to nothing less than a monumental case of scientific fraud (Wotiz, Rudofsky 84; Wotiz 93). We are not in a position to comment on the authenticity question, nor does it make much difference for our purpose whether the insight actually did take place, and under what circumstances. Since the story is conceivable and resembles countless other cases of anecdotal evidence regarding insights that were instrumental in creative acts of problem solving, we shall accept it on the basis of its plausibility.

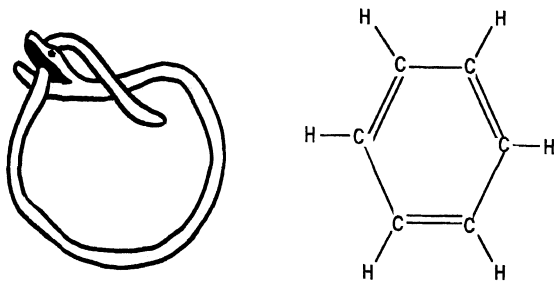

Figure 1: Kekule's insight and discovery of the structure of the benzene molecule

The story is repeated in many scientific discussions and is used to illustrate theories of scientific discovery. Interestingly, different authors utilize the Kekulé story to advance different arguments. In particular, it serves as either an example of mental visual imagery (e.g. Finke 90; Miller 86; Perkins 8 1; Shepard 78), or as an instance of analogical mapping and transfer (e.g. Ariety 76; Holyoak, Thagard 95; Langley, Jones 88; Martindale 95; Peterson 93). Robert Weisberg, a prominent psychologist, appears to be undecided, or else he sees no problem in citing the case alternately as an example of imagistic reasoning on

one occasion, and analogical reasoning at another opportunity (Weisberg 86, 88).

Proposition: The dual significance of the Kekulé insight, or shall we say anecdote, helps bring forth the question we are concerned with: what is the relationship between visual imagery and visual analogy? In particular, we are interested in the use of imagery and analogy in reasoning, especially in processes of creative problem solving. Are imagistic reasoning and analogical reasoning two separate processes, that may or may not occur in tandem? If Kekulé's case can indeed be seen as both a case of imagery and a case of analogy, there are three possible relationships among these two cognitive phenomena, leading to the following alternative propositions: a) two distinct operations occurred in sequence: first imagery, then analogy b) two distinct operations coincided: systems that activate imagery and systems that activate analogy operated concurrently. c) a single imagistic system operated; one or more subsystems activated imagery in the specific tasks of mapping and transfer that we refer to as visual analogy. We propose that the third proposition is most convincing, provided we accept a wide enough definition of imagery. We shall substantiate this claim through the use of an example from the domain of design, where both imagistic reasoning and analogical reasoning are quite frequent.[3]

2.2 The Dommelsch Advertisement Campaign

Nico van Scherpenzeel is a senior art director in a large Dutch advertising company that was commissioned to design an advertisement campaign for Dommelsch beer. It was to be multi~functional (posters and magazines, television and radio. Here we focus on the posters only) and was meant for the summer months (91). The design process description that follows is based on several conversations with van Scherpenzeel, who was in charge of the endeavor, approximately six months after the campaign was launched.

The design process can be divided into four phases. In phase 1 the designers looked for a theme. After a long search during which many ideas were considered and discarded, they decided to capitalize on the fact that it was summer time and chose to compare having Dommelsch beer with going abroad on vacation. The theme was encapsulated in the slogan: "But we have Dommelsch." van Scherpenzeel testifies that this decision and the slogan that goes with it came in a flash. In phase 2 a prototype poster was developed, envisaged as containing a picture (photograph) of a desirable item, such as a beautiful car or house, next to which (to the right) the thematic slogan was to be printed out. Phase 3 still contained one

[3] We do not yet possess empirical evidence to support our claim that visual analogy is a special case of visual imagery; however, it should be possible to run neuroccognitive experiments using fMRI, PET or other imaging devices, to inspect the respective areas of the brain that are activated during visual analogy and visual imagery processes.

picture, but instead of just any item, the picture was to show a famous building or site in a popular vacation destination. The slogan was now to be printed below the picture. In phase 4 the final idea was crystallized: van Scherpenzeel refers to it as picture-rhyme. The single picture was replaced by two "rhyming" pictures: one depicting the tourist attraction (as in phase 3), and another (to the right, or beneath) showing Dommelsch beer (bottle, glass, crate, etc.). The slogan is added as before.

Figure 2: The 'Great Wall of China' Dommelsch advertisement [4]

First three posters were produced, to be followed later by three more. The production of the

[4]Advertisement by Noordervliet and Winninghoff/Leo Burnett, Amsterdam (Nico van Scherpenzeel, Art Director) for Dommelsch Bierbrouwerij B.V. Reproduced with permission.

series followed the usual working procedure in advertisement: the site-photographs were mostly stock items supplied by an agency according to specifications that took into account the resemblance to a Dommelsch bottle (or glass, etc.). The sites included the Statue of Liberty, the guards of Buckingham Palace, the tower of Pisa, the Eiffel tower, the Niagara Falls and the Great Wall of China. We will use the latter poster, shown in Figure 2, as an exemplar. First an attractive photo of a section of the China Wall was chosen. Then a complicated operation took place to fabricate its Dommelsch analog: two photos were taken: one of a flat landscape, the other of a configuration of beer crates, on a flat surface. The two images were fed into a computer where they were combined and manipulated to create a composite image. In the final picture beer crates appear to be arranged in a hilly landscape, such that the geometry of both the beer crate construction and the landscape features bear a striking resemblance to the Wall of China and the landscape in which it is set. The China photo was selected from amongst several photos of the Wall that were supplied for inspection as per request. van Scherpenzeel testifies that the idea of the crate-arrangement flashed the minute he had first contemplated the particular photo that was subsequently selected.

2.3 Process Reconstruction and Analysis

What does the story of the Dommelsch poster design have in common with the insight that led to the discovery of the benzene molecule structure? We propose that the inventive processes in the two cases (different as they might be in magnitude and significance) were similarly served by analogical reasoning executed through mechanisms of imagery. In both cases the individuals involved had been working on their respective problems for some time: deciphering the structure of a molecule, and designing an effective advertisement campaign for beer. Both men were experienced thinkers in their domains. Kekulé had of course dealt with molecule-structures for a long time, and had considered many alternative configurations for the benzene molecule, before and after his insight. Hexagonal configurations in plane were one such possibility (Gruber 95). Kekulé used imagistic reasoning as a matter of routine; in fact he "had develped a special form of imagery in which he could think directly; he had already had many ideas in that modality, and now he had a new one." (Gruber ibid. 410). Likewise, van Scherpenzeel was no newcomer to the design of advertisement campaigns, and by the time he was ready for the last poster in the Dommelsch series, he had already successfully established the "picture-rhyme" strategy. Both men were committed to their work, and thought about their tasks continuously, conducting active searches for possible solutions while using their personal versions of the tools of their trades.

The tools of the trade deserve our special attention, because they include consulting pictorial material of various kinds. van Scherpenzeel retrieved photographs related to the task from an external database, although he did not and could not know how many pictures

he might have to inspect before one that could be of use would be found. Kekulé, who did not have a database at his disposal, nor had he advanced his search to the point where he could explicitly specify what type of pictures he would be interested in inspecting, resorted to the "special form of imagery" that he was practiced in. He knew from past experience that if he allowed himself to generate seemingly random images, the preoccupation with a specific problem would most probably place constraints on those images such that only relevant images would be formed. Alternatively, relevant images would attract his attention sufficiently to notice their relevance (compare to Chase and Simon's findings that expert chess players avoid trial and error and to a large extent are able to make successful moves only) (Chase, Simon 73). Kekulé's account contains no information regarding any other images he might have entertained during the long search related to his problem solving efforts. It is safe to assume, however, that he had had other images that may have faded away leaving no trace because he could not identify any helpful clues in them, hence expelling them to the realm of irrelevancy. Thus both individuals inspected pictures, externally or internally, with the explicit intention of finding in them hints that could potentially advance their search efforts. (Compare these 'hunting' episodes with a report by de Bono on the development of a medical device to test lung function. de Bono deliberately decided to wander around a large dime-store because it was "a setting full of the sort of objects which might trigger an idea. It was just a matter of wandering around until something suggested itself' (de Bono 71, 111).

Relevant pictures were found. Kekulé "saw" twisting snakelike rows, one of which "had seized hold of its own tail", thus forming a ring. van Scherpenzeel saw a vivid composition depicting three stretches of the wall with three watch-towers, placed where the wall changes direction. Why were these pictures useful? Because our problem-solvers could form images of analogous pictures, that provided the basis for solutions to their problems. They did not simply identify analogies: they created them by imaging a target configuration. In Kekulé's case - a ring of carbon atoms; in van Scherpenzeel's case - a configuration of beer bottle crates laid out in a landscape. Kekulé's target image was a model that needed testing and evaluating. van Scherpenzeel's target image had to be fabricated in reality to resemble the inner picture (using photographic and computational techniques). With hindsight it is easy to identify the analogy between the source and target pictures in each case. But the processes of mapping and transfer from source to target are not so obvious and indeed, they must have been two-way processes, such that source to target moves were accompanied by target to source moves as well. In other words, the source is not merely retrieved from memory; it may be retrieved and transformed, or actually reconstructed and reorganized almost from scratch until its representation is suggestive enough so that mapping and transfer to a candidate target that is being constructed becomes possible[5]. In this view, neither source nor target are stable

[5]A case in point that illustrates the role of transformation in retrieved information is the erroneous representation of a famous building plan, in the process of resolving and architecural problem. The

representations until the process of transfer has been completed: both representations are susceptible to transformations in order to strengthen and verify the accuracy of mapping.

Two points must be emphasized. First, tasks that are aimed at creating novel objects and entities (two and three-dimensional), such as various design assignments or scientific modeling that is expressed in diagrams etc., naturally require visual or pictorial representations. Inner representations through imagery are known to complement external representations of various sorts. It is therefore not surprising that individuals who solve problems of this type make use, extensive use at times, of their imagistic powers. Second, we must remember that there are differences between well-structured problems and illstructured problems, such as the design and scientific discovery problems we are concerned with. In solving the latter a search is indispensable, as most ill-structured problems have more than one possible solution. When an ill-structured problem also falls into the 'visual' category, then all kinds of configurations may provide bases for a representation that would be considered a satisficing (to use Herbert Simon's terminology) solution to the problem. Using all available strategies, the problem solver is likely to conduct a search in which pictorial representations, external and internal, are generated and transformed.

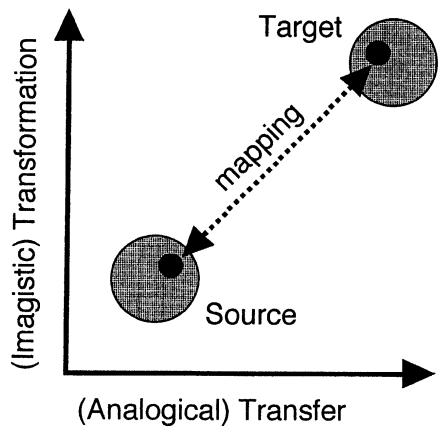

Figure 3: Diagram of dynamic imagistic/analogical search process in creative problem solving

This explains the importance of the dynamic principle we outlined above, which allows for the entertainment of unstable, possibly vague images, that may undergo serious

error permitted the architect to form a visual analogy that helped him to resolve his problem (Goldschmidt, 1994; 1995).

transformations throughout the process of transfer between candidate-source and candidatetarget. Figure 3 is a diagram of the dynamic, two-way mapping process in a creative search involving imagistic transformation and analogical transfer.

3 Interactive Imagery

If image generation is a very complex cognitive phenomenon, imagistic reasoning must involve even more complex cognitive structures. Kosslyn (94) counts four image generation types classified according to the mechanisms that trigger them, and differentiating between arrangements of image components and the individual components themselves. Image components can be arranged using either categorical or coordinate representations of spatial relations, and individual components can be imaged by allocating attention or by retrieving and activating visual memories. In imagistic processes involved in problem solving such as we have described, images are formed that were never before perceived: the designer, the inventor and the discoverer work to represent something new, a brainchild that has never previously been stored in memory and can therefore not be retrieved. Kosslyn explains this possibility as follows: "People can "mentally draw," visualizing patterns that they have never seen and that are not simply new combinations of familiar components. This process is a special case of the attention-based imagery..." (Kosslyn 94, 3 10). Elsewhere Kosslyn adds that this special type of imagery utilizes image transformation, i.e. the ability to modify patterns in images (Kosslyn 99). Imagery based on attention implies interpretative processes and indeed, Finke (90) has shown that by manipulating given components (selected randomly from a restricted set) in imagery, people can successfully combine them into useful objects when interpretative constraints are imposed (the object is to belong to a specified category, say furniture). Finke found that not only could people perform such tasks fast and arrive at results, but that the exercise yielded unique objects that were often rated as creative by judges who evaluated them. So convincing were the findings regarding the power of imagery, that Helstrup and Anderson set forth to show that various decomposition tasks can be performed in imagery alone as accurately as when subjects were also allowed to sketch in the process (Helstrup and Anderson 91).

However, in Helstrup and Anderson (ibid.) and in subsequent studies of mental synthesis, sketching experience was not a variable and subjects where not experienced sketchers. Versijnen et al. (98), who were familiar with designers' propensity for sketching, found those results counter intuitive. They were able to show that in composition tasks that resemble Finke's experiments, creativity scores favor sketchers over subjects who used only imagery, when the subjects were experienced sketchers and when their manipulation of the given components included restructuring. When the subjects were not experienced sketchers, no significant differences in creativity were found between those who sketched and those who used only imagery, as in Helstrup and Anderson (ibid.). Sketching skills are therefore belived to have the power to amplify imagery, when exercised.

We relate these results here because they appear to support the claim that imagery can be ‘stretched’. In other words, depending on task requirements and an individual's abilities, a larger or smaller number of imagistic systems or subsystems may be activated. If a composition task can benefit from external representation and if the individual is fluent in the production of such representations (no cost or low cost in terms of attention), then we can expect sketching to be a contributing factor. This is certainly true for design tasks in the hands of skilled designers. Goldschmidt (91) has called the act of sketching in the early phases of conceptual design 'interactive imagery', meaning the simultaneous or almost simultaneous production of a display and the generation of an image that it triggers." (ibid. 131). Interactive imagery enables the designer to converse with his materials, to use Schbn's terminology, in a dynamic manner, taking advantage of the speed with which images and sketches can be generated and transformed.

In some respects the use of visual analogy in creative processes serves a purpose similar to that of sketching: a problem-solver faced with a complex ill-structured problem, in a domain that requires visual representation, recruits available resources that can fortify his or her search strategies. Since no rules are at hand to reason with, the immediate goal is to generate visual representations that are rich with clues that may potentially lead to a solution by way of similarity. There are differences between the use of an external support tool such as sketching and the spontaneous (for the most part. but not always) search for and identification of analogous situations, which is not necessarily accompanied by an external representation. But there is no contradiction between these strategies which are not mutually exclusive, and may actually be practiced concurrently, as the representations in question are external and internal respectively and orthogonal to one another.

Sketching can, of course, be practiced independent of imagery (and for purposes other than inventive problem solving). Likewise, an analogy can be identified without the use of imagery. However, in a creative search analogies are not just identified, they are created as a result of the manipulation and transformation of images - as we have seen earlier. Outside of the laboratory, it is very rare for a problem-solver to be presented with two displays that can be identified as analogous. It is almost always the case that the problem-solver contemplates one display (externally or internally). and forms an image of another display that is identified as analogous to the first. As noted earlier, it is of no consequence which of these displays is the source and which is the target. Images undergo transformations, because the problem-solver uses them to reason about the problem at hand. The transformations bring the two images close enough together to allow mapping and transfer, and this is true whether we subscribe to a theory of structural relations or a theory of functional multiconstraints in analogy (e.g. Gentner 83; Gick and Holyoak 80; Holyoak, Thagard 95; Keane 88). These are the arguments that drive us to believe that visual analogy is a cognitive operation that can only occur when imagery is activated. Seeing it as a

subsystem of imagistic mechanisms makes it possible to posit that the analogy subsystem is activated when a complex problem requires amplification of the basic imagistic system. It is reasonable to assume that this subsystem is more developed in some individuals, who are therefore likely to exploit it more effectively. This explains the relative rareness of spontaneous appeal to analogy in general problem solving, and its successful utilization by certain specialists.

Like imagistic reasoning, which turns out to be an elaborate ability in which we can identify a number of types, analogical reasoning is also a complex cognitive capability. We distinguish between surface and deep analogies, and between within-domain and betweendomain analogies or, to be even more elaborate, between local, regional and long-distance analogies (Dunbar 95). Due to historical circumstances, the study of imagery and analogy have developed as two distinct research fields that have little in common. This may be in part due to the fact that analogy researchers seem to have taken for granted the fact that so many instances of analogy involve visual representations. In reported instances of the use of analogy in creative problem solving, researchers remain indifferent to the question of whether these representations are external or internal. Elsewhere, the effort to model analogical reasoning computationally has concentrated on the identification of analogies rather than on their generation. The primarily visual nature of the information to undergo mapping and transfer, and the fact that it is not always simply perceived but may actually have to be generated first, have never received ample attention (e.g. Vosniadou and Ortony 89).

4 Visual Analogy and the Acquisition of Design Expertise

While we know what a decisive role analogical reasoning has played in many a discovery, invention and creative achievement, we also have substantial evidence that spontaneous recourse to analogy as a reasoning strategy in problem solving is quite uncommon, to say the least. Many known instances of beneficial use of analogy involve visual images, and we may therefore assume that the habit of "thinking in pic&ures" (in adulthood) may actually lead to a better chance of employing analogical reasoning in problem solving. It is therefore reasonable to hypothesize that problem-solvers will perform better if made to use visual images, and in particular, if made to use analogy when it is doubtful that they would do so spontaneously. Should this prove to be the case, we may want to ask what implications for learning in general does the induced use of analogy have, if any? And what specific implications can we identify for learning to design?

Schunn and Dunbar (96) have shown that in science, priming enhances access to previously acquired knowledge, which can be and is used to solve current problems, among others through analogical mapping and transfer. This process is implicit and in an experiment, subjects were not aware of what made them think of the particular knowledge they were

able to elicit in order to successfully solve a novel problem. As we have already reported above, Gentner and Medina (98) who conducted experiments involving the use of analogy with children, have come to comparable conclusions. They stress that not only do subjects perform better when they had previously solved analogous problems, but that long-term learning can be shown to occur in which abstract concepts are better understood, abstracted, and applied in subsequent appropriate situations, (Note the difference between these results and those obtained by Gick and Holyoak, where problem solving was not enhanced following passive exposure of subjects to analogous situations. See section 2).

Casakin and Goldschmidt (Casakin 97; Casakin and Goldschmidt 99, Casakin and Goldschmidt 00) asked whether the use of analogy would produce similar effects in design problem solving, and set out to investigate this question empirically. Three groups of subjects with varying design experience participated in the experiments: experienced architects, advanced architecture students and novice architecture students. Each subject solved a number of well-defined (one correct solution) and ill-defined (any number of acceptable solutions) design problems that were presented under three conditions: a) The design problem was administered with no additional material or commentary; b) The design problem was administered while the subject was also shown a panel with visual displays (approximately two dozen images), some of which could be used as source analogs for the problem, and some that could not be related to it in any way. The latter served as 'noise , 6 ; c) Same as b, but subjects were also explicitly encouraged to identify relevant images among the displays and use them as source analogs in their designing. The subjects' solutions to the design problems were expressed in sketch form, on one or more sheets of paper. These sketch-designs were assessed by three naIve judges (all experienced architects) for arrival at a solution in the case of well-defined problems (yes/no scores), and for the quality of the solutions and their creative merits on a scale of 5 points in the case of ill-defined problems (for the purpose of this study we do not differentiate between wellfilldefined, and welllill-structured problems). As the degree of agreement among the judges was very high, the results were seen as perfectly reliable and were consequently submitted to statistical analyses. Details of the experimental settings and the results are reported elsewhere (Casakin, ibid; Casakin and Goldschmidt, ibid.); here we want to highlight some of the findings which, we believe, are highly relevant to our study of analogy as a reasoning strategy in designing, and in learning to design.

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[6]For example, one of the problems called for the design of a typical floor of a prison, such that the guards maintain visual control over the doors of all 80 cells; each cell was to have at least one exterior wall. Some of the images in the panel of displays that accompanied the presentation of this problem described the following: a wheel (with spokes); a clock, a top view of a round slide projector carousel; the logo of the Shell company; a fan; a plan of a fortified city; a number of floor plans of residential buildings with a central core (elevators etc.) and radial 'fingers' in which apartments are located.

We choose to distinguish between experimental results obtained for problems termed 'well-defined' and those termed 'ill-defined'. For well-defined problems, architects were assisted by the presence of visual displays but instructions to use analogy had no significant effect on the quality of their work. In contrast, students' performance was enhanced by instructions to use analogy, whereas the mere presence of visual displays had no significant improving effect on their work. For ill-defined problems, on the other hand, the pattern of results was different. Both architects and students obtained significantly higher scores when using visual displays, and giving them instructions to use analogy maximized their designquality scores. These results are summarized in Table One (based on findings reported in Casakin 97. Analysis in Casakin and Goldschmidt 99 and Casakin, Goldschmidt 00 pertains only to ill-defined problems). Our interpretation of these findings follows below.

Table 1: Design quality scores, novice and experienced designers under different problem solving conditions

		Novice designers (students)				Expert designers (architects)	
		Well-defined		Ill-defined		Well-defined	Ill-defined
		Begin.	Advan.	Begin.	Advan.		
Baseline (control)	No display provided No instructions to use analogy	--	--	2.227	2.58	--	2.809
Design quality improvement	Displays provided No instructions to use analogy	p<.469	p<.541	2.621 p<.039	2.939 p<.092	p<.091	3.236 p<.046
Additional design quality improvement	Displays provided Instructions to use analogy given	p<.063	p<.042	3.463 p<.001	3.731 p<.002	p<.148	3.984 p<.001

For experienced professionals, well-defined problems are not novel and the architects in our experiments could solve them using routine, low-level strategies. The visual displays they contemplated, when they did, where enough to evoke similar problems they had solved in

the past, and possibly even straight-forward algorithms they could easily access and activate for a successful solution of these problems. Therefore, no recourse to analogy, a higher-level strategy, was necessary: cognitive systems are 'economically minded' and normally prefer to activate the lowest-level strategy possible for any given task. For students, however, with much less experience, even these well-defined problems were new and were therefore treated as ill-defined, at least to a certain extent. A low-level strategy such as contemplating visual displays was not sufficient and therefore when they could enlist analogy, which they did only when they were instructed to do so, it helped them solve the problem.

In contrast the ill-defined, or ill-structured problems were novel to all subjects and therefore analogy came in handy as a high-level design problem solving strategy. These results confirm our hypotheses regarding the role of analogy in the solving of ill-structured problems. But we still have to address those findings that pertain to the impact, or lack thereof, of visual displays on the effectiveness of problem solving in our experiments. The results we have obtained may seem confusing and inconsistent, but we think they are not. In fact, we believe they provide real insight into visual thinking and its relationship with analogical reasoning.

We postulate that the effect of visual displays on problem solving mechanisms is variable in strength and usability, in accordance with the mode of processing the displays undergo. For the sake of simplicity we will talk about two modes of processing, although there might be, and probably is, a more flexible continuum. At the lower-level mode, visual displays serve as pointers, icon-like memory-aids that facilitate retrieval of information from memory. When that information is stable and securely stored in memory retrieval is easy and this simple aid may be sufficient to promote successful integration of the retrieved information in the process of problem solving. When, however, knowledge is not available, or is not very securely stored and therefore difficult to access, the icon-like reminders are of no avail unless the system is induced to process them in the second, higher-level mode. This is where imagistic reasoning (as opposed to perception) comes into play and possibly interactive imagistic reasoning only (we have not thecked possible correlation with factors such as sketching). We must stress that this mode of processing, which requires more cognitive resources, is reserved only for those cases where lower-level strategies are insufficiently effective. If correct, this theory explains the results we have obtained. In the case of ill-structured problems all our subjects had to resort to the higher-level processing mode of the visual displays and indeed, in order to exploit the visual images which they understood to harbor potentially useful information, or clues, for beneficial moves. In the case of well-structured problems the two groups of subjects were in very different positions as problem-solvers. The experienced designers required only low-level processing modes, and the mere provision of visual displays was sufficient. The novices (both beginning and advanced students) were unable to make use of this economical low-level processing mode,

and therefore contemplating the displays had no beneficiary effect. Only when forced to go into higher-level processing modes as a result of the requirement to use analogy, could they take advantage of the availability of the displays.

5 In Conclusion

The implications for design education are clear to us. People can be trained to maximize the processing resources they are endowed with. Novices can be shown (e.g. through feedback) how the use of a certain cognitive strategy, analogic reasoning in this case, can be beneficial to them. At the same time repeated use can help stabilize domain-knowledge in their memory such that in subsequent cases it might require less effort, i.e. a lower-level processing mechanism, to retrieve and activate that knowledge[7]. Students of analogy have shown that the process of analogic mapping and transfer is a two-way operation: from a known example to abstraction, and from abstraction to a new candidate-example in the realm of the problem situation in hand (e.g. DeJong 89; Goldschmidt 95; above - sections 2 and 3). Novices who exercise analogical reasoning eventually appropriate the abstractions, implicitly or explicitly, thereby gaining in knowledge and in the ability to apply it, as claimed by Gentner and Medina (98). The use of analogy therefore brings similarity-based and rule-based reasoning very close together, and enables them to interact successfully in problem solving. In this interaction reasoning by similarity is typically helpful as a triggering device in the early phases of a search for ideas, whereas reasoning by rules follows suite in subsequent phases of development, refinement and evaluation. An example in the context of architecture is Le Corbusier's design for the roof of the Ronchamp chapel. According to the architect's own testimony (Broadbent 73) the shape of the roof was first conceived as a simile of a crab's shell which had been lying on his desk for quite some time. Le Corbusier, who made frequent use of analogical reasoning, transferred the structural principle of the double-membrane shell to the roof. Later, however, adjustments had to be made to ensure the proper functioning of the new concrete shell to its role as a roof, taking into account insulation, drainage and so on, as well as aesthetic and structural properties concerning the very large overhangs that lend the building its special silhouette.

In the various design domains, where typically ill-structured problems limit the serviceabilty of rules in the process of problem solving, reasoning by visual analogy is undoubtedly a very important strategy. For novice designers, it is also a most beneficial strategy for learning to design, and in particular for the acquisition of new concepts.

[7]The use of between-domain analogies is much harder, if not impossible to train for. Their spontaneous elicitation in problem solving appears to be all but reserved to extraordinarily creative thinkers. Chupin (98) reports a pedagogic experiment in which students of architecture are given between-domain analogies with which they are required to work. Unfortunately the report does not give any information on the effect of this experimental procedure on students' performance.

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